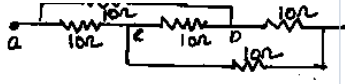
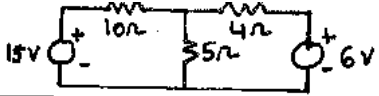
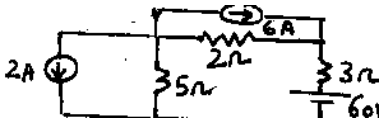
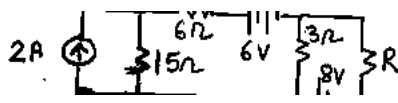
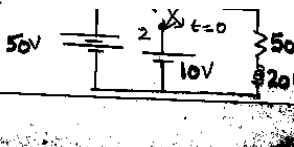


MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

First Semester, September, 2025

Paper Code: ES-107**Subject: Electrical Science****Time: 1½ Hrs.****Max. Marks: 30****Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.**

Q. No.	Question	Max. Marks	CO(s)
1.(a)	Calculate the equivalent resistance between the terminals a and b as shown in network. 	2.5	CO 1
1.(b)	Define the following with examples: (i) Active and Passive elements (ii) Linear and Non-linear elements	2.5	CO 1
1.(c)	Define the following terms: (i) Apparent and True power (ii) Reactive Power and Power Factor with suitable formulas.	2.5	CO 2
1.(d)	An alternating current varying sinusoidally with a frequency 50Hz has an rms value of 10A. Write down the expression for the instantaneous value and find its value at 0.0125s after passing through a positive maximum value.	2.5	CO 2
2.(a)	State Maximum Power Transfer Theorem for DC network. Derive the condition of maximum power transferred to load and also find the value of load resistance.	5	CO 1
2.(b)	Calculate the current in 4 Ω resistor in the circuit using Thevenin Theorem. 	5	CO 1
3.(a)	Derive RMS value, Average value, Peak factor and Form factor of Half wave rectified alternating current.	5	CO 2
3.(b)	Find the current in 5 Ω resistor using Superposition Theorem. 	5	CO 1
4.(a)	Find the value of R to have maximum power transfer in the circuit. Also find the maximum power. 	5	CO 1
4.(b)	The switch has been in position 1 for a long time. It is moved to 2 at t = 0 . Obtain the expression for i_L(t) for t > 0 . 	5	CO 1

**** End of Question Paper ****